

Internal flow within RNA Condensates

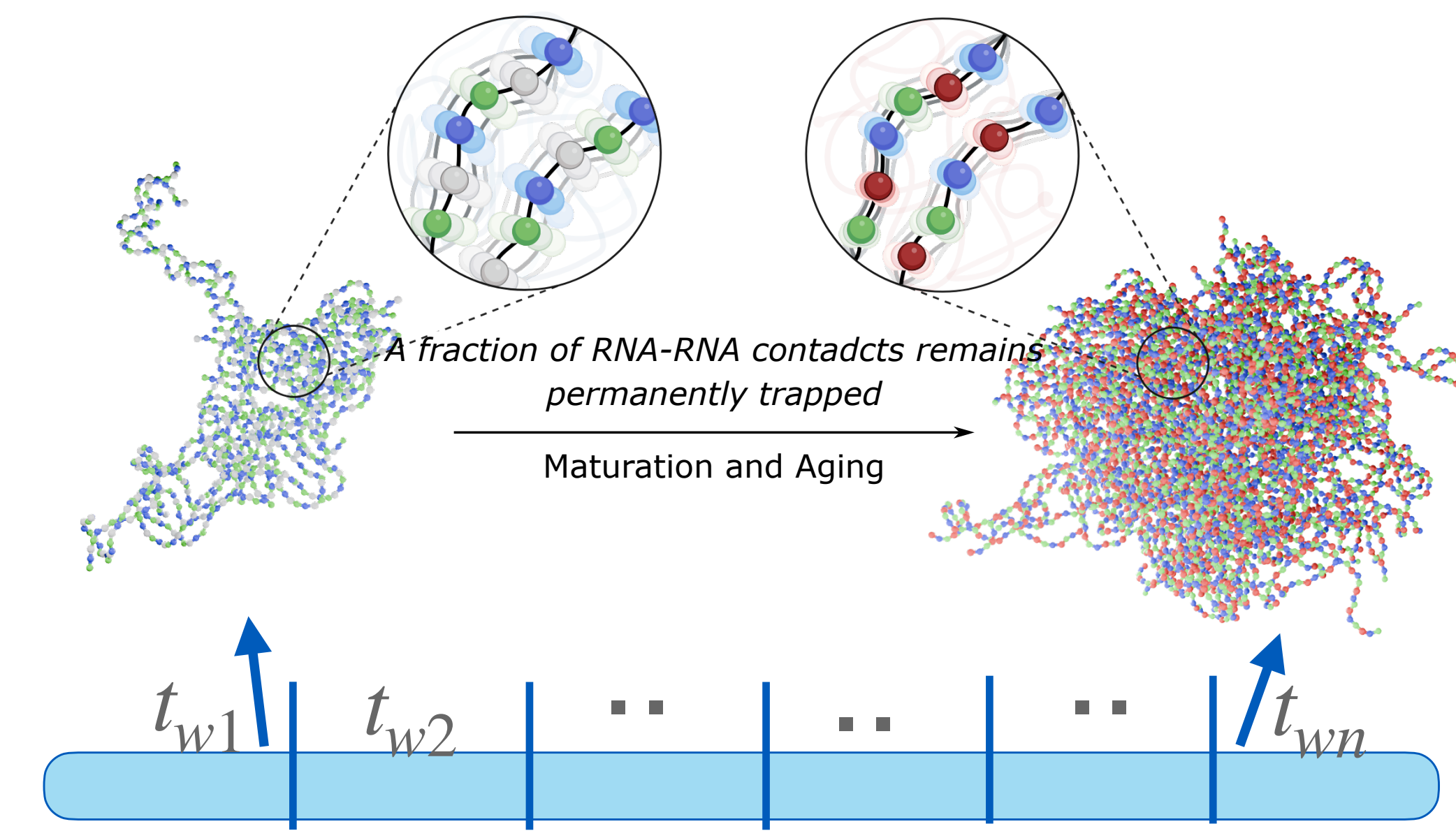
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Introduction

- RNA condensates formed by homotypic RNA–RNA interactions are implicated in neurodegenerative disease (Huntington's, ALS, FTD).
- Formed by liquid–liquid phase separation, they mature from fluid droplets into viscoelastic, gel-like states, with slowed diffusion and growing internal heterogeneity.
- Using coarse-grained MD + an Eulerian voxel velocity-field analysis, we map the internal flow of a single condensate.



Methods

- Single-Interaction-Site (SIS) model and Langevin dynamics to simulate multi-CAG31 RNA (64 chains).
- Pairwise mean-square displacement (pMSD) analysis to study the droplet dynamics,

$$\langle \Delta R_{ij}^2(\tau) \rangle = \left\langle \frac{1}{N_p} \sum_{i,j} \Delta R_{ij}^2(\tau) \right\rangle$$

Where $\Delta R_{ij}(t_1, t_2) = R_{ij}(t_2) - R_{ij}(t_1)$ and N_p is number of eligible pairs.

To find the diffusion coefficient, we fit power-law to pMSD as,

$$\langle \Delta R_{ij}^2(\tau) \rangle = D(t_w) \tau^\kappa = D_0 t_w^{-\alpha_w} \tau^\kappa$$

Coarse-grained velocity field averaging

MD Trajectory (n frames, N beads, Positions $r_i(t)$)

Per bead, per time origin t_0 (fixed lag $\Delta\tau$)

$$v_i = \Delta r_i(t_0 \rightarrow t_0 + \Delta\tau) / \Delta\tau$$

Spatial binning into voxels
Voxel index $\text{idx} = r_i / 5\text{\AA}$ (COM frame) and accumulate $\sum v_i$

$$\text{averaging over origins } t_0; \text{ mean voxel velocity } \langle \vec{v} \rangle = \frac{\sum v_i}{n(\Delta\tau)}$$

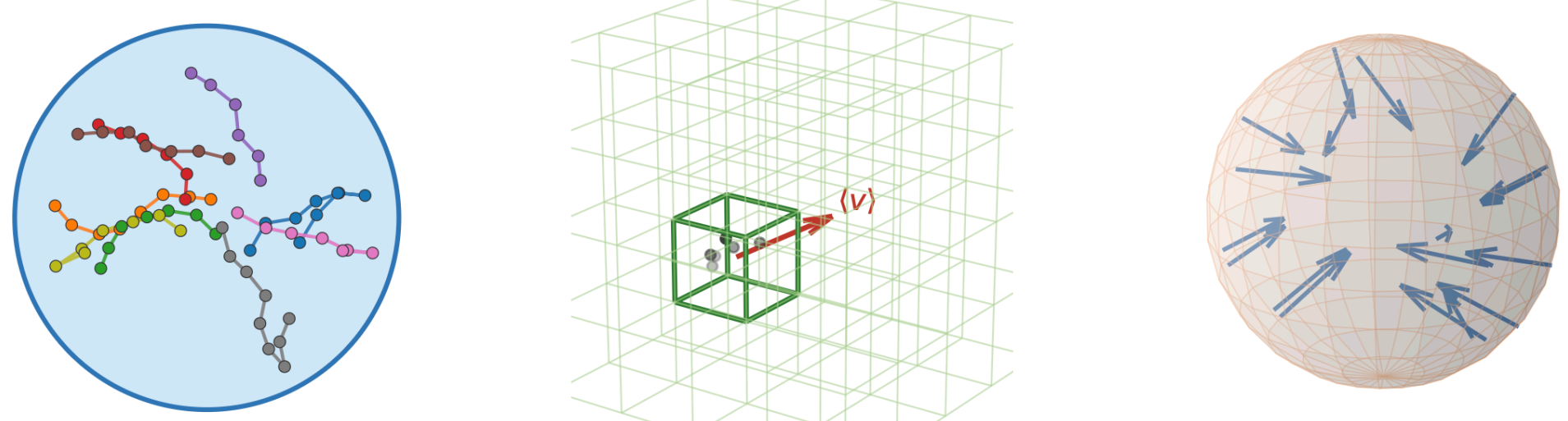
Field analysis: divergence $\vec{\nabla} \cdot \vec{v}$, vorticity $\vec{\nabla} \times \vec{v}$, deformation.

Schematic representations

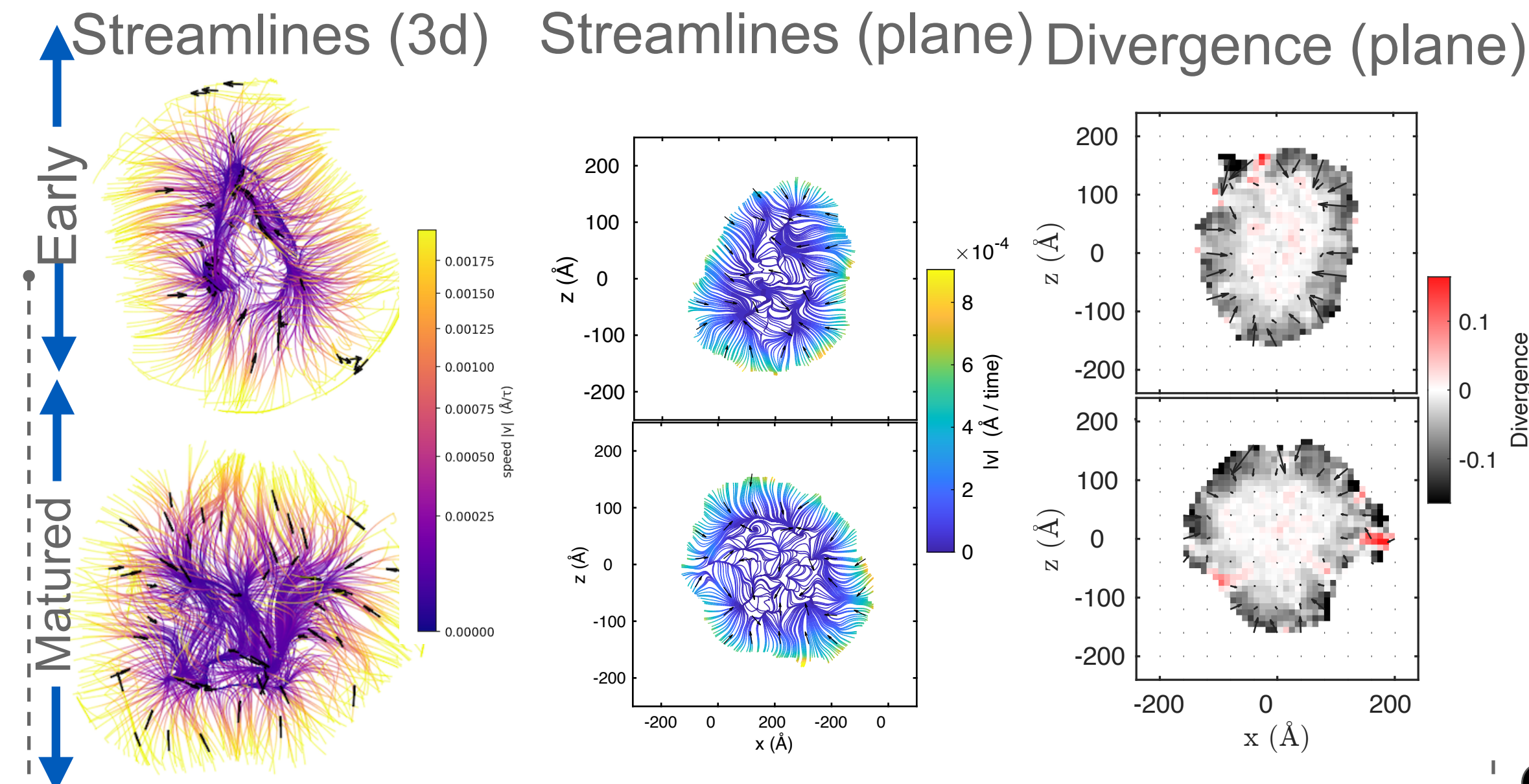
(a) nucleotides

(b) bin + average in voxel

(c) mean velocity field



A Internal velocity distribution

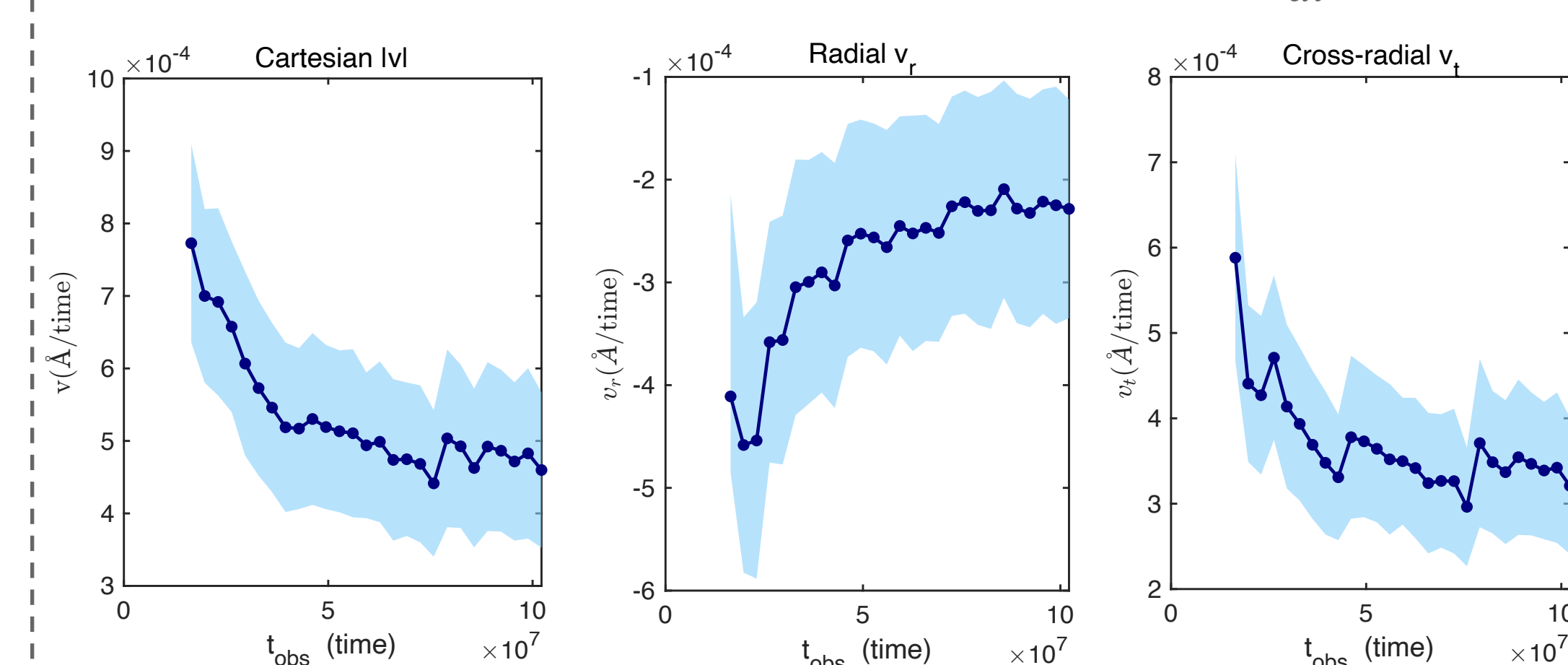


- Early: one central sink; matured: many peripheral sinks.** Early on, chains flow inward and converge toward a single sink. With age the interior becomes inaccessible (crowding), and convergence fragments into multiple sinks near the surface.

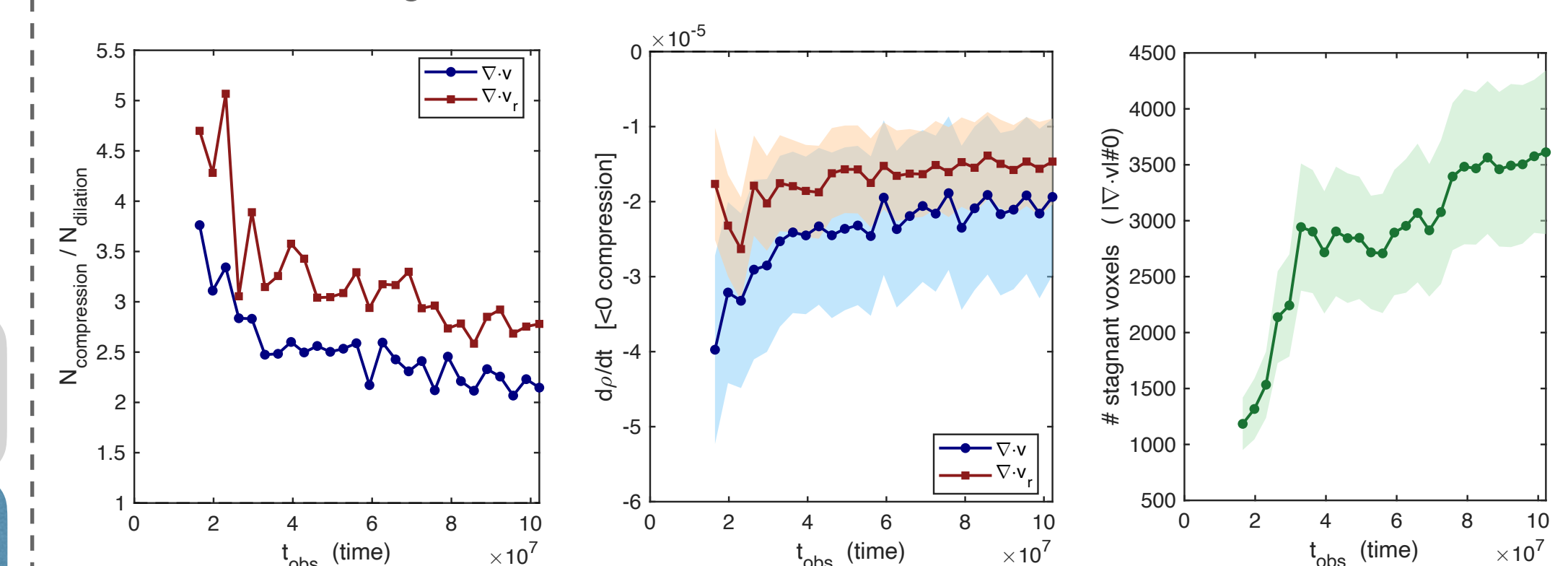
- Divergence (compression) is likewise strongest and inward at the surface, and weakens as the condensate matures.

From, continuity equation,

$$\vec{\nabla} \cdot \vec{v} = -\frac{d\rho}{dt}$$

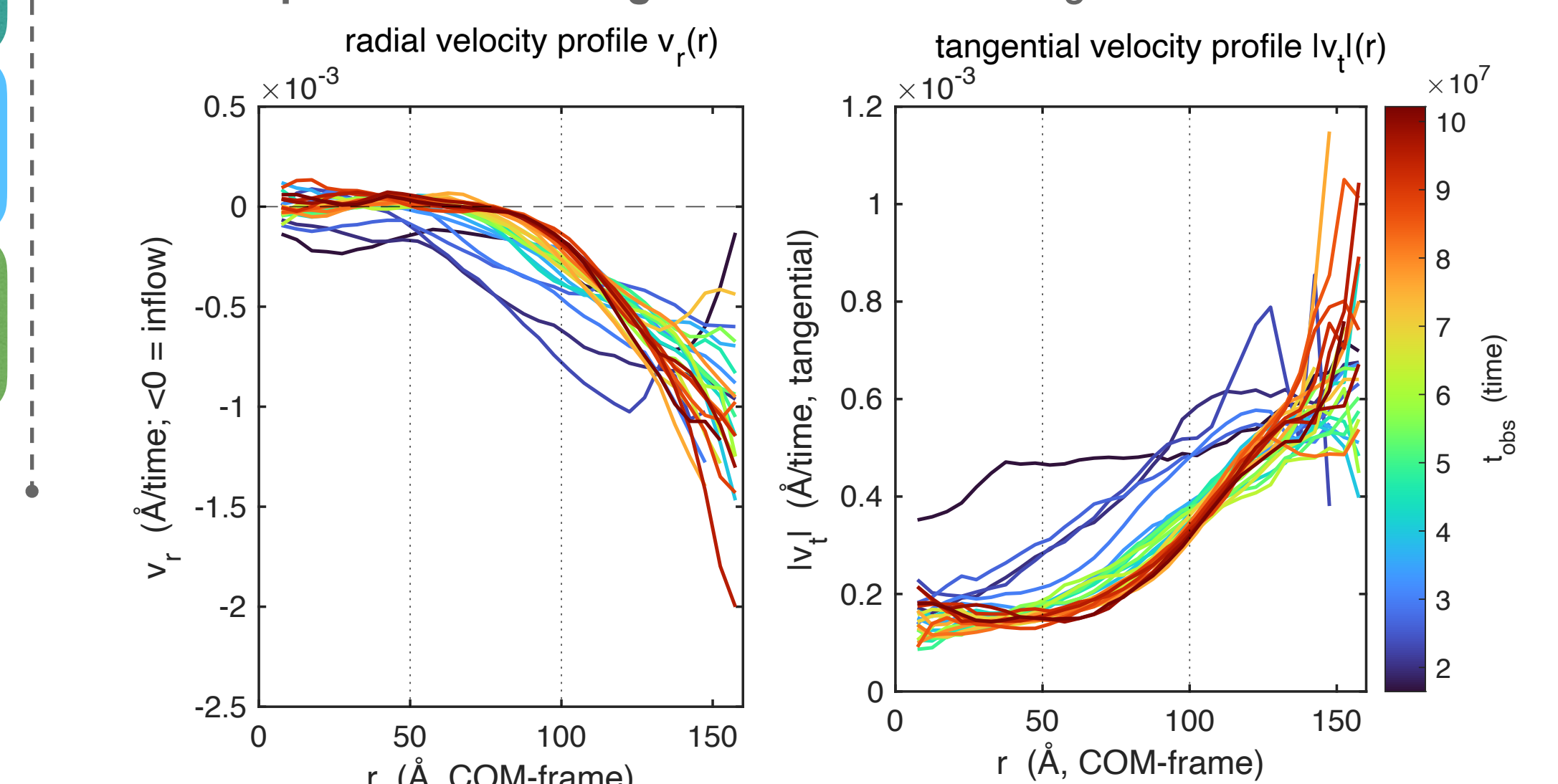


- Speed slows with age:** Cartesian $|v|$, Radial inflow and tangential flow weakens with age.



- Compression dominates, then slows down:** ratio $\gg 1$, with age $\rightarrow 1$

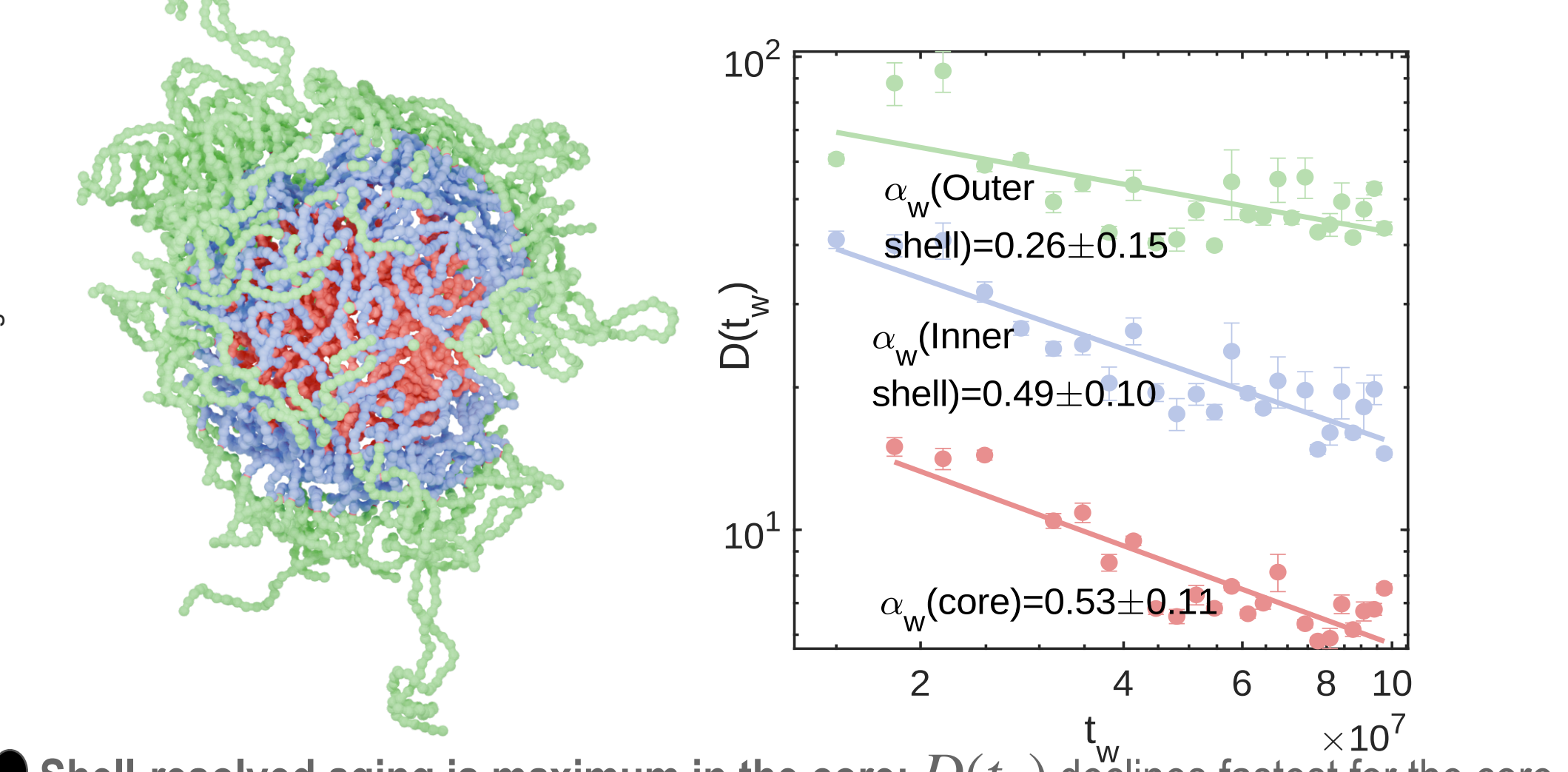
- The droplet freezes in regions:** near zero divergence voxels increases



Radial profile: quiet core, contracting surface. $v_r \approx 0$ in the core and dives negative toward the surface, steepening with age.

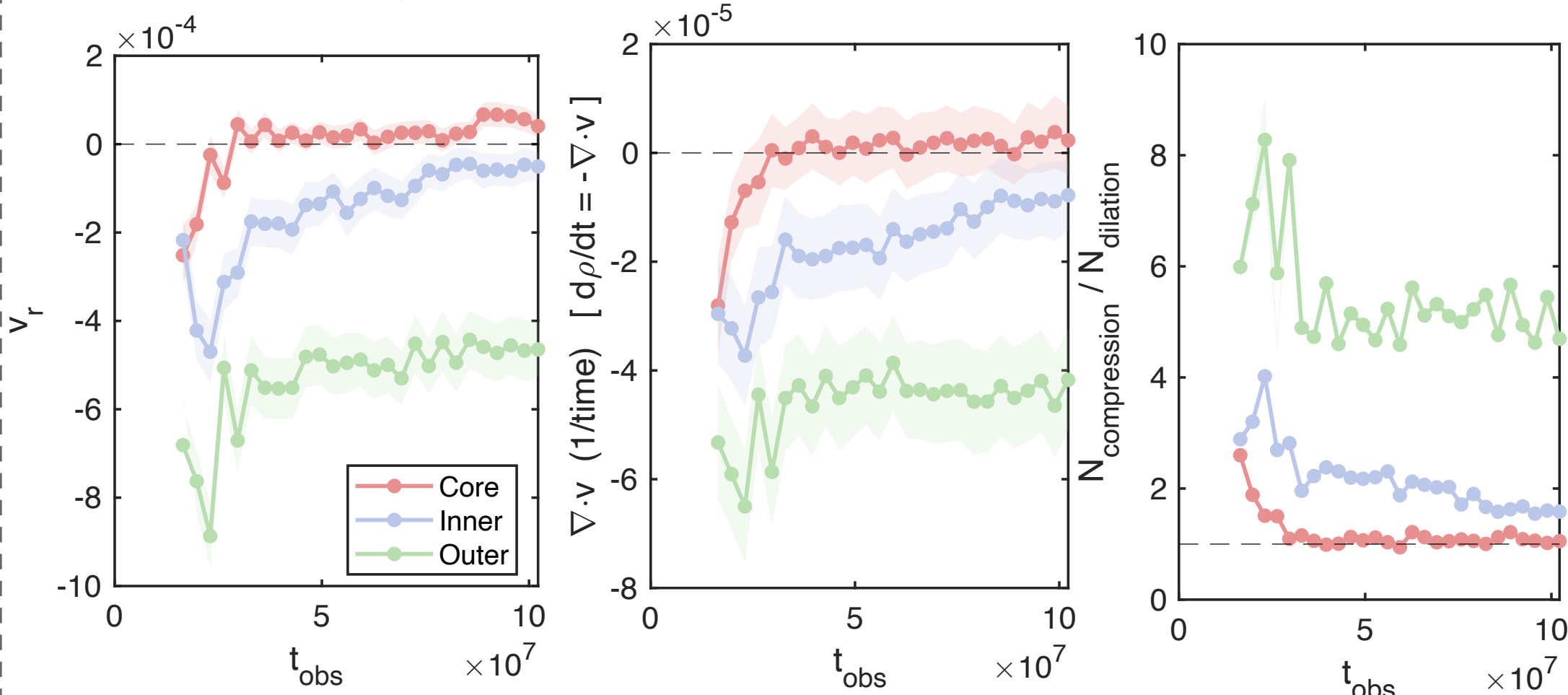
B Spatial and temporal heterogeneity

To explore the spatial heterogeneity, we analyse the droplet concentric shell wise from the COM of the droplet (a).



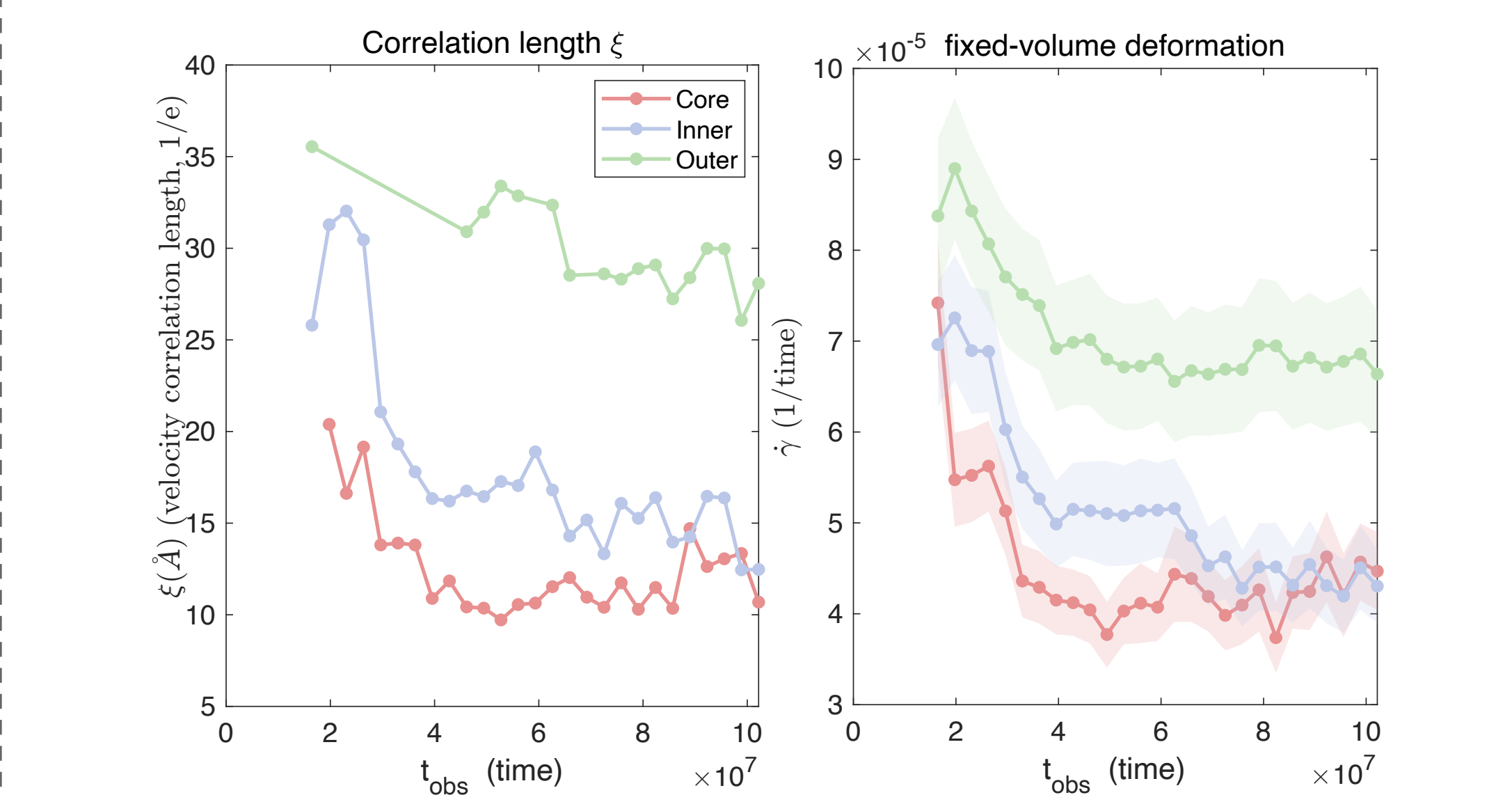
- Shell-resolved aging is maximum in the core:** $D(t_w)$ declines fastest for the core, slowest for the outer shell

- aging exponent α (core > inner > outer)



- The chains at outer region converges rapidly towards interior where the core becomes saturated by incoming chains and no more converging.

- The ratio also depicts that compression is dominated in the outer region.

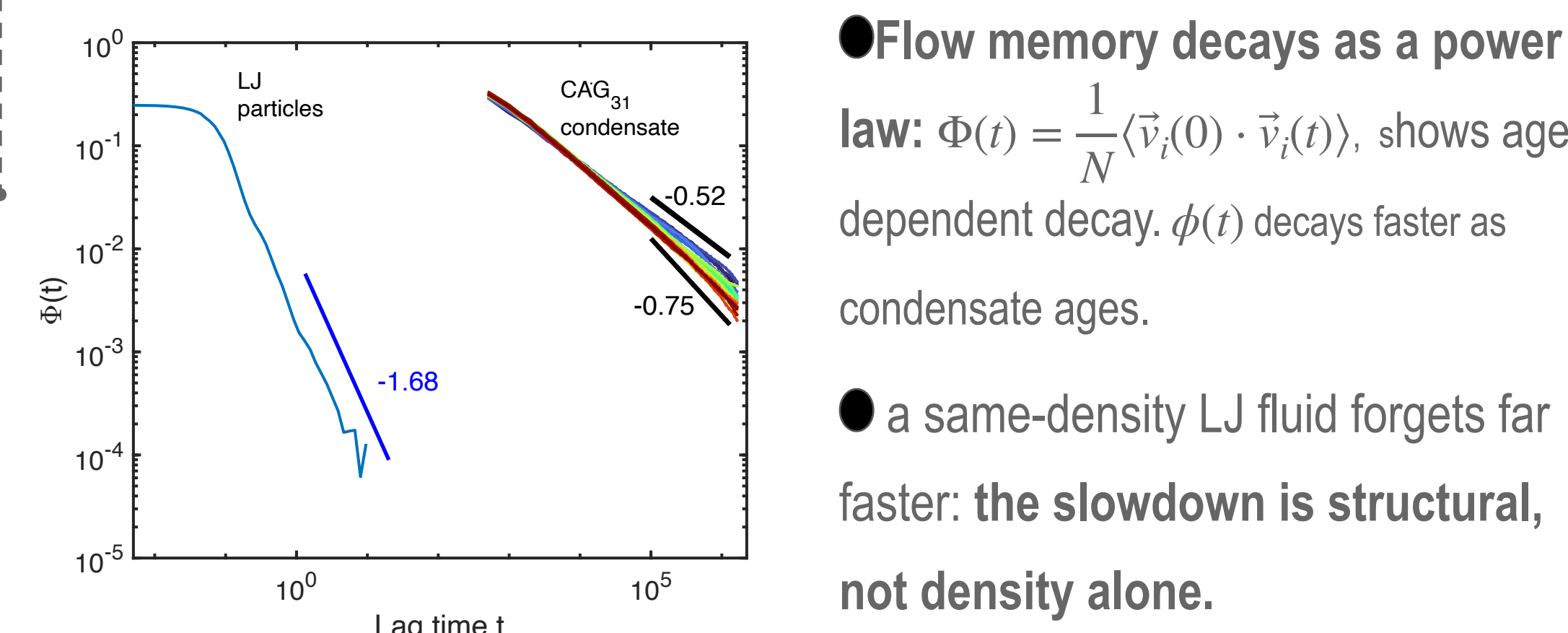


$$C(r) = \frac{\langle \vec{v}(\mathbf{x}) \cdot \vec{v}(\mathbf{x} + \mathbf{r}) \rangle}{\langle |\vec{v}(\mathbf{x})|^2 \rangle}, \text{ with } \frac{C(\xi)}{C(0)} = e^{-1}, \text{ Coherence is lost with age.}$$

$\xi(1/e)$ shrinks over time: fastest in the core

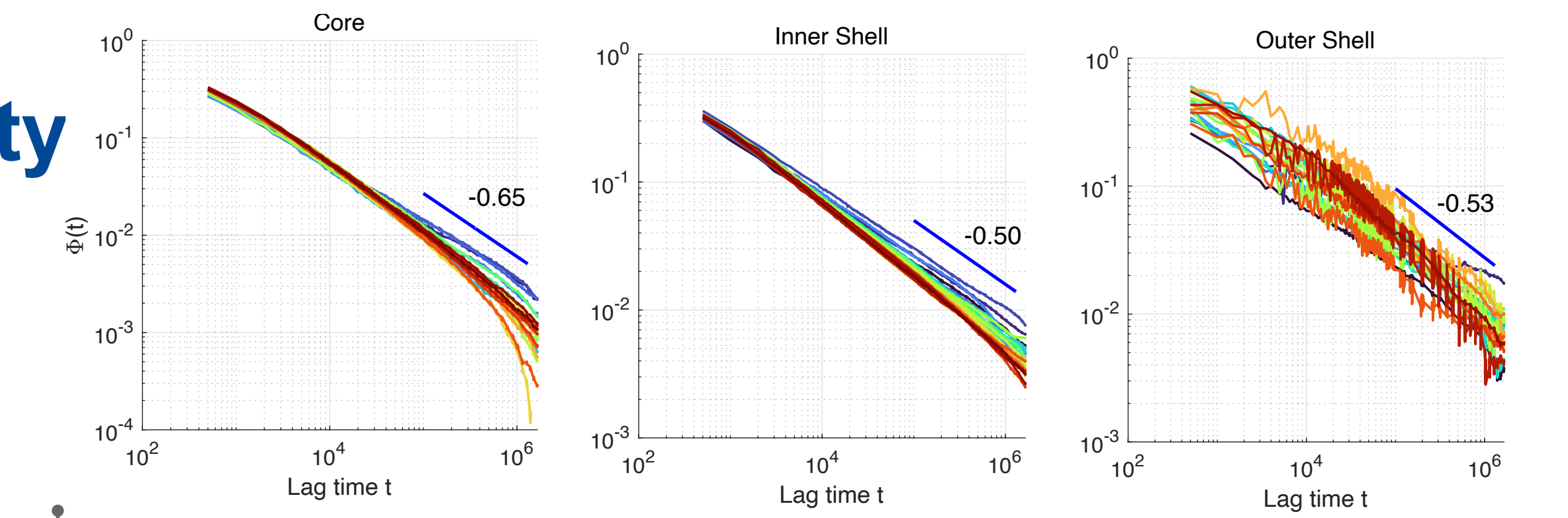
$$\dot{\gamma} = \sqrt{2 E_{ij}^{\text{dev}} E_{ij}^{\text{dev}}}, \quad E_{ij}^{\text{dev}} = \frac{1}{2} (\partial_j v_i + \partial_i v_j) - \frac{1}{3} (\nabla \cdot \mathbf{v}) \delta_{ij}, \text{ the surface keeps shearing/rearranging while the core is the most arrested against shape change.}$$

C Dynamical relaxation of condensate



- Flow memory decays as a power law:** $\Phi(t) = \frac{1}{N} \langle \vec{v}_i(t) \cdot \vec{v}_i(t) \rangle$, shows age-dependent decay. $\phi(t)$ decays faster as condensate ages.

- a same-density LJ fluid forgets far faster: **the slowdown is structural, not density alone.**

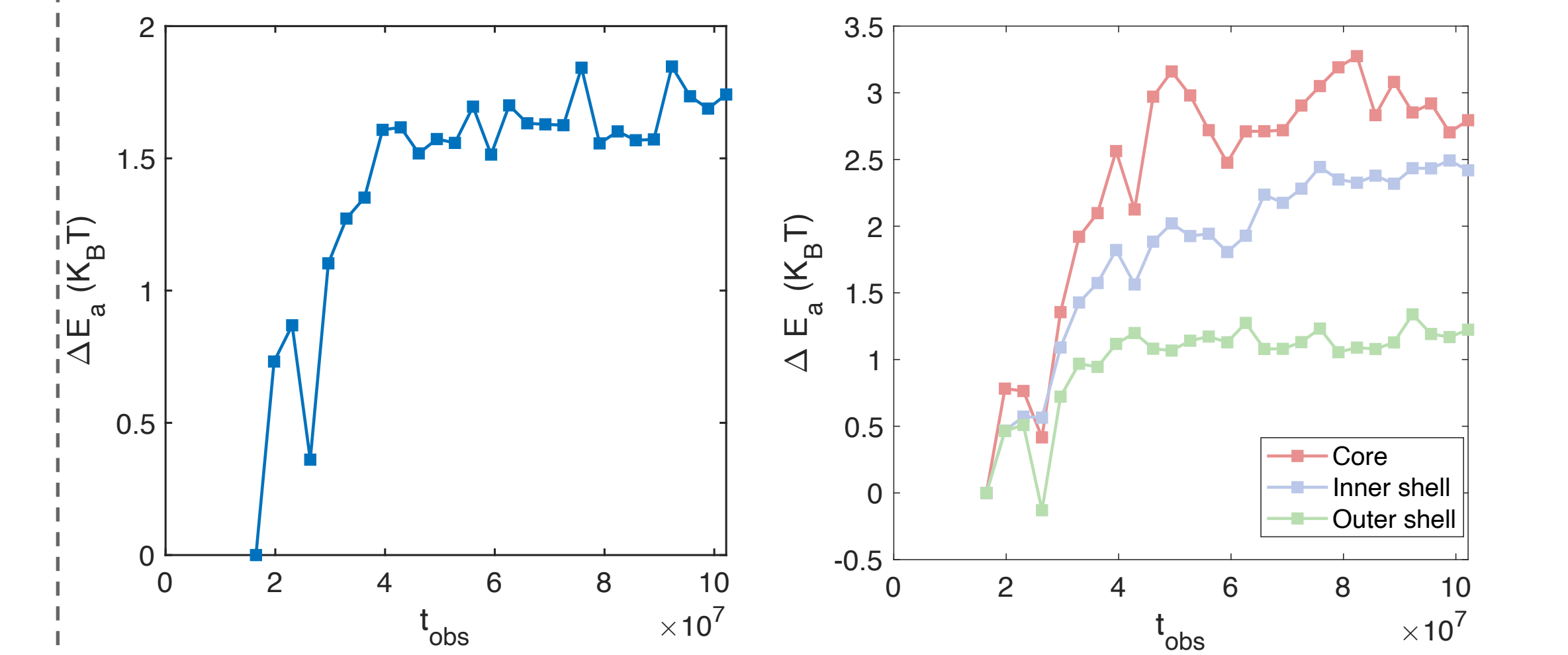


- Relaxation is spatially heterogeneous:** RNA chains loses flow memory faster within core.

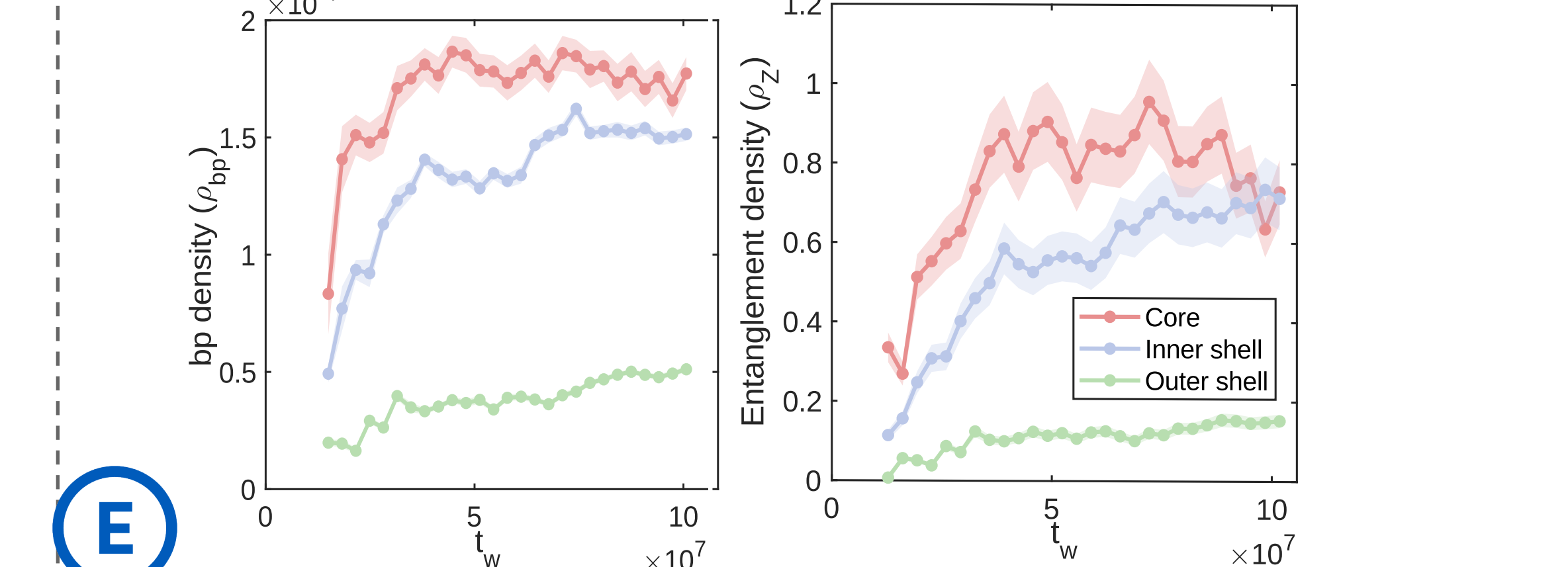
D Molecular origin

- Following Arrhenius/transition form for structural relaxation, $\tau_\alpha = \tau_0 \exp(E_A/k_B T)$, we calculate $\Delta E_A(t_{\text{obs}}) = k_B T \ln \frac{\tau_\alpha(t_{\text{obs}})}{\tau_\alpha(t_{\text{ref}})}$.

- The reconfiguration barrier grows with age and Barrier is highest in the core**

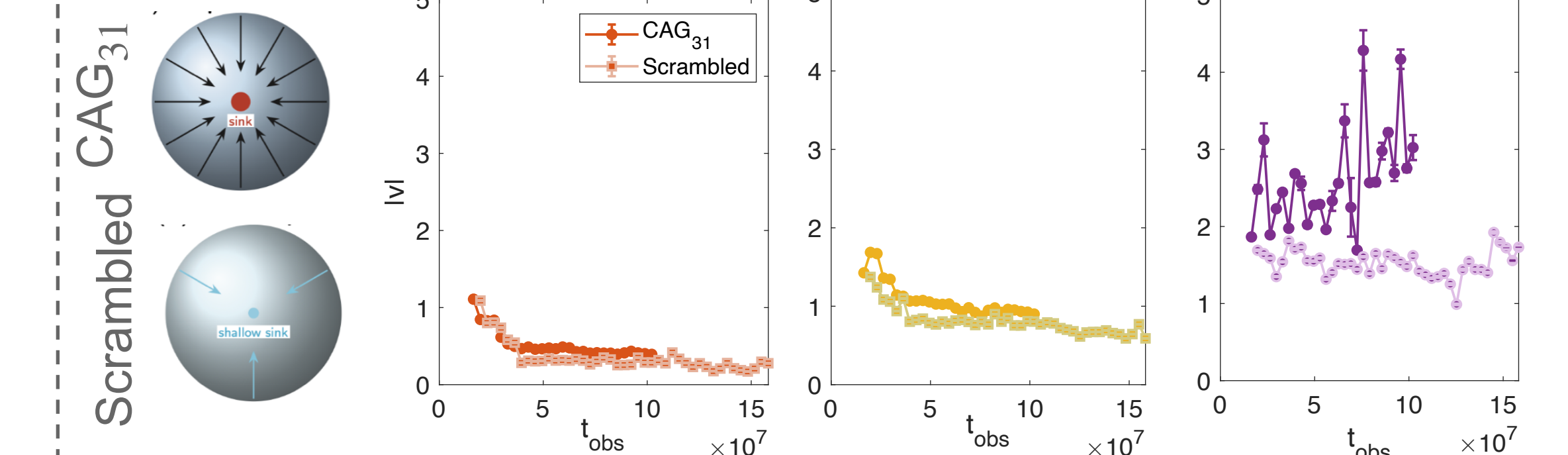


- Condensate dynamics is base-pairing network and entanglement driven:** RNAs near the core are more entangled, forming a dense network driven by enhanced base pairing.

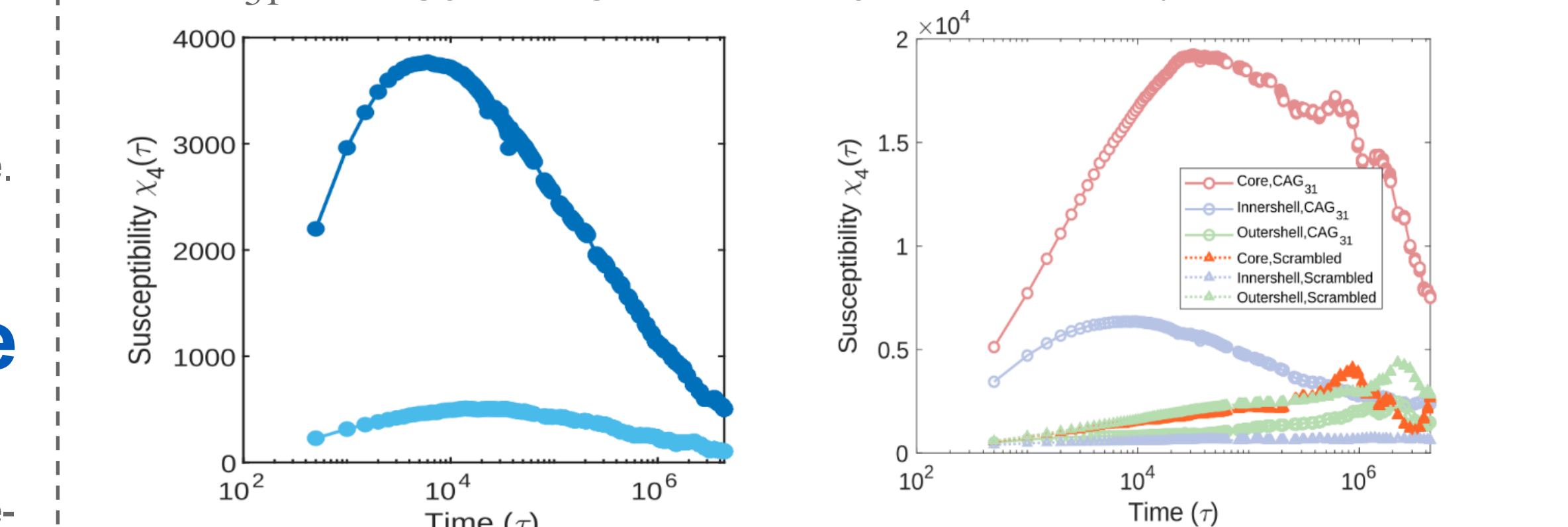


E Sequence effect in flow and dynamics

Sequence controls the slowdown: In every shell, intact CAG31 sustains more directed flow



- CAG31 is strongly heterogeneous:** undergoes pronounced dynamical variations



Conclusion

- The condensate undergoes a universal compaction that intensifies at the surface and saturates rapidly in the core.
- The dynamical slowdown is strongly sequence-dependent: scrambled RNAs of identical composition show no pronounced aging transition.